AP CALCULUS AB - UNIT 1

# 1.10 Exploring Types of Discontinuities

## Lesson Overview

- Classify removable, jump, and infinite discontinuities using one-sided evidence.
- Distinguish the three named types from oscillatory or domain-gap failures.
- Use factor cancellation and multiplicity to classify rational-function discontinuities.
- Write the limit evidence before naming the type.

## Key Knowledge and Vocabulary

**Removable.** A finite common limit exists, but the function is missing there or has the wrong value.

**Jump.** Both one-sided limits are finite and unequal.

**Infinite.** At least one one-sided limit is $+\infty$ or $-\infty$.

**Oscillatory.** No single output destination exists and the behavior is not simply unbounded.

**Classification protocol.** Compute left limit, right limit, two-sided conclusion, then compare with the point value.

## Explanation and Strategy

**Step 1.** For rational functions, factor first: complete cancellation creates a hole; a remaining denominator factor creates a pole.
**Step 2.** For piecewise functions, evaluate the branch limits at every breakpoint.
**Step 3.** Do not call every DNE limit a jump; identify the actual one-sided fingerprint.

## Solved Examples

### Worked Example: Removable rational discontinuity

Classify $f(x) = \frac{x^2-9}{x-3}$ at $x = 3$.

**Solution.** For $x \neq 3$, $f(x) = x + 3$, so the common finite limit is 6 while the original function is undefined. The discontinuity is removable.

### Worked Example: Jump in a piecewise function

$g(x) = x + 1$ for $x < 2$ and $g(x) = x^2$ for $x \geq 2$. Classify at 2.

**Solution.** The left limit is 3 and the right limit is 4. They are finite and unequal, so the discontinuity is a jump.

### Worked Example: Infinite by multiplicity

Classify $h(x) = \frac{x+1}{(x-4)^2}$ at $x = 4$.

**Solution.** The numerator is positive near 4 and the denominator is positive and tends to 0. Both sides tend to $+\infty$, so the discontinuity is infinite.

## Leveled Practice

### Shared Representation / Data

| Fingerprint | Classification |
|---|---|
| Finite common limit; point missing or wrong | Removable |
| Finite unequal one-sided limits | Jump |
| At least one unbounded side | Infinite |
| No finite/unbounded destination due to persistent oscillation | Other nonexistence |

### Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

**1.** Classify $\frac{x^2-16}{x-4}$ at $x = 4$.

**2.** A graph has finite one-sided limits 2 and 5 at $x = 1$. Classify.

**3.** A function tends to $+\infty$ on both sides of $x = -3$. Classify.

**4.** A function oscillates between $-1$ and 1 near 0 without shrinking. Is this removable, jump, or infinite?

### Medium Level

Apply the idea in one or two connected steps.

**5.** Classify every discontinuity of $f(x) = \frac{(x-3)(x+2)}{(x-3)(x-1)}$.

**6.** Classify $g(x) = \frac{x+2}{(x+2)^3}$ at $x = -2$.

**7.** Classify the greatest-integer function $\lfloor x \rfloor$ at an integer $n$.

**8.** Classify $\frac{|x|}{x}$ at 0.

## Hard Level

Connect representations, conditions, and justification.

**9.** Classify all discontinuities of $\frac{x^2-1}{(x-1)^2(x+3)}$.

**10.** For $f(x) = \tan x$, classify the discontinuity at $x = \pi/2$.

**11.** A function has $\lim_{x\to 2} f(x) = 7$ and $f(2) = 0$. Classify and state the repair.

**12.** A formula is undefined on the entire interval $(1, 2)$, but defined on both sides. Is the gap at $x = 1.5$ a removable point discontinuity?

## Difficult Level

Use nonroutine structure and precise mathematical reasoning.

**13.** Classify $f(x) = \frac{(x-a)^m}{(x-a)^n}$ at $x = a$ for positive integers $m, n$.

**14.** Can a rational function have a jump discontinuity? Justify.

**15.** Classify $x\sin(1/x)$ at 0 when no value is assigned there.

**Challenge Level**

Synthesize, construct, generalize, or prove.

**16.** Construct one function having a removable discontinuity at $-1$, a jump at 2, and an infinite discontinuity at 4.

**17.** Prove that changing a function at a single point can repair only a removable discontinuity.

## AP-Style Practice

### Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

**1.** Which evidence identifies a jump discontinuity at $x = c$?

**(A)** Both sides approach the same finite value.
**(B)** Finite one-sided limits exist but are unequal.
**(C)** Both sides approach $+\infty$.
**(D)** The function value is missing.

**2.** For $f(x) = \frac{(x-2)^2}{(x-2)^3}$, the discontinuity at 2 is

**(A)** removable
**(B)** jump
**(C)** infinite
**(D)** none

**3.** Persistent oscillation near a target is classified as

**(A)** always removable
**(B)** always jump
**(C)** always infinite
**(D)** another form of limit nonexistence

### Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

**4.** A calculator graph of $\frac{x^2-4}{x^2-x-2}$ seems smooth except near $x = -1$ and $x = 2$. The correct classification is

**(A)** holes at both points
**(B)** poles at both points
**(C)** a pole at $-1$ and a hole at 2
**(D)** a hole at $-1$ and a pole at 2

**5.** A table near $x = 0$ alternates close to $-1$ and 1 as inputs shrink. The discontinuity is most likely

**(A)** removable
**(B)** jump
**(C)** infinite
**(D)** oscillatory

### Free-Response Practice – No Calculator

Let $f(x) = \dfrac{(x-1)(x+4)}{(x-1)(x-3)^2}$.

**(a)** Identify every excluded input. [1 points]

**(b)** Classify the discontinuity at each excluded input. [3 points]

**(c)** State the removable-hole limit and the signs of the infinite limits at 3. [3 points]

## Free-Response Practice – Calculator Required

A calculator table for $g$ near $x = 5$ gives left values 1.9, 1.99, 1.999 and right values $-2.1$, $-2.01$, $-2.001$. The point value is $g(5) = 10$.

**(a)** Estimate both one-sided limits. [2 points]

**(b)** Classify the discontinuity. [2 points]

**(c)** Can changing $g(5)$ make $g$ continuous? Explain. [2 points]